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Xihai Luo

Data Scientist / Software Developer

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Ambitious Master of Artificial Intelligence Design & Development graduate student seeking a Data Analyst, Data Scientist, or ML Engineer WIL/Co-op (Sept 2026 – April 2027). I offer hands on experience in time series forecasting, predictive modeling, and frontend development to drive data driven decision making and scalable AI solutions.

SKILLS

Tools and Languages	Python, SQL, Pandas, NumPy, Scikit-learn, Jupyter Notebooks
Data Science	Exploratory Data Analysis (EDA), Feature Engineering, Data Wrangling, SHAP
Machine Learning	Regression, Classification, Cross Validation, Probabilistic Models, Clustering

PROJECTS

Canadian Labour Demand Forecasting January 2026 — April 2026
Seneca Polytechnic Toronto, Ontario, Canada

- Engineered a time series forecasting model to predict labour demand in the Accommodation & Food Services sector over a 5 month horizon, utilizing Job Vacancy counts from Statistic Canada and Job Bank posting (2023-2025) as proxy variables
- Developed and tuned a suite of 5 machine learning models including XGBoost, LightGBM, and Random Forest, against a SARIMAX baseline, with the **Random Forest model achieving high accuracy MAPE score of 4.72%** on the test set
- Synthesized key insights from Exploratory Data Analysis (EDA) into a visual presentation for policymakers, demonstrating a sector wide demand decline and providing strategic recommendations to prioritize employee retention over aggressive hiring

TECHNICAL EXPERIENCE

Frontend Developer April 2025 — August 2025
Polaron Solar Energy Toronto, Ontario, Canada

- Collaborated closely with UX designers and product managers to create a scalable design system, built with React + Tailwind CSS, documented and tested in Storybook, which improved developer workflow and design consistency
- Optimized web and mobile performance by implementing lazy loading, CDN caching, and dependency analysis, which **improved Core Web Vitals (CLS, LCP, FCP) by 50%**
- Refactored our CI/CD pipeline by setting up a staging environment on Alibaba Cloud ECS and building custom deployment scripts in GitHub Actions, **reducing deployment times by 40 minutes**

Application Developer August 2023 — April 2025
Ministry of Public and Business Service Delivery and Procurement Toronto, Ontario, Canada

- Architected the Ontario Design System (ODS) React component library using TypeScript, which was **adopted by 4 cross-functional teams to streamline government application development**
- Redesigned critical government portals and features, such as the postal code lookup, using Figma and AODA/WCAG 2.0 standards to ensure full accessibility compliance and improved usability.

Web Developer July 2022 — September 2023
SilverPonies Web Inc. Vancouver, British Columbia, Canada

- Led performance optimization initiatives for high revenue Shopify and WordPress clients, resulting in a **20% increase in page performance through Core Web Vitals analysis**
- Delivered a comprehensive homepage redesign as Lead Developer, which **doubled loading speeds** and directly contributed to improved customer retention
- Developed a suite of reusable UI features and WordPress plugins that enhanced user engagement and streamlined the user experience for multi-million dollar client accounts

EDUCATION

Master of Artificial Intelligence Design and Development (MAI), Seneca Polytechnic January 2026 — Present
Winter 2026 President's Honour List - CGPA 4.0/4.0

Bachelor of Applied Science and Engineering, University of Toronto Completed July 2020
Major in Electrical Engineering
Minor in Sustainable Engineering

ACTIVITIES

Scientist Farzand Husain Education and Welfare Foundation Canada (SFHEW), Robotics Instructor 2025 — Present